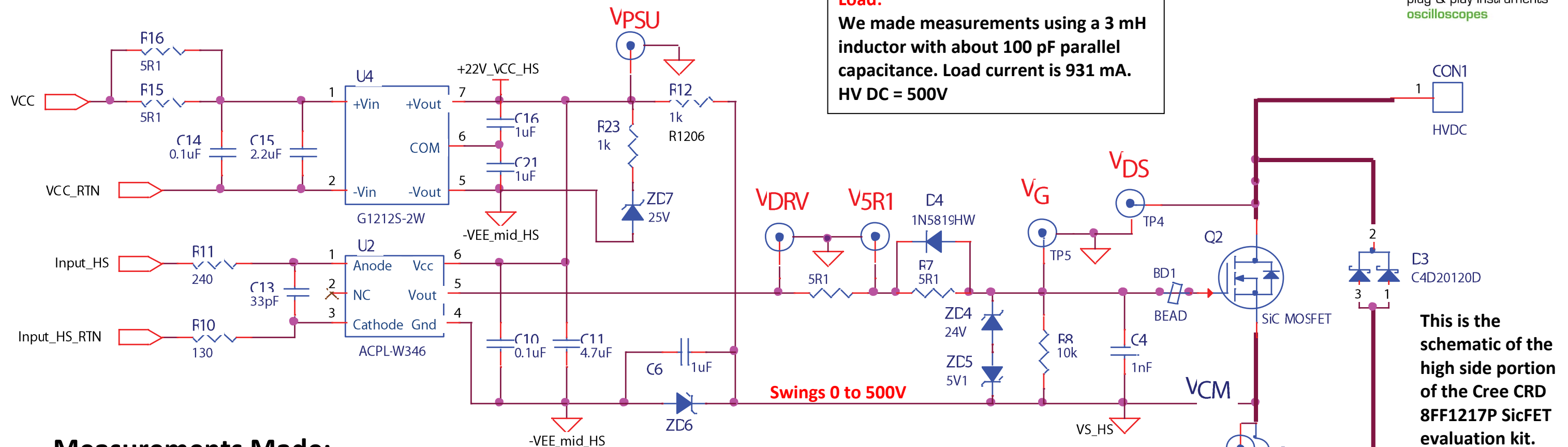


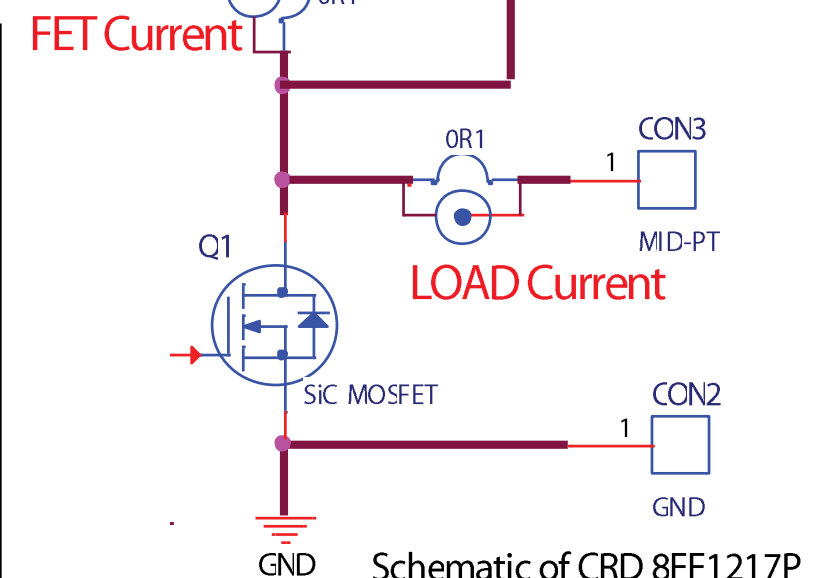
Cree SicFET High Side Measurements using the CS448



Measurements Made:

- **Gate Voltage**
Measure V_G
- **Gate Charge**
Measure V_{DRV} and V_{5R1} .
Current $I = (V_{DRV} - V_{5R1})/5.1$.
Charge $Q = \int I dt$
- **Gate PSU Voltage**
Measure V_{PSU}
- **Conduction Power Loss**
Measure V_{DS} with V_{CE-SAT} Probe,
Measure I_{DS} with **FET Current**
Power = $V_{DS} * I_{DS}$ in W
Energy = $\int \text{Power } dt$, = J per switch cycle.
Use V_G to identify ON period
- **Switching Power Loss**
Measure V_{DS}
Measure I_{DS} with **FET Current**
Power = $V_{DS} * I_{DS}$ in W
Energy = $\int \text{Power } dt$, = Joule per switch.
Multiply Joule per switch by switch frequency to estimate total Power Loss
Use V_G to identify switch period
- **Diode Loss**
Measure V_{DS}
Measure I_{Diode} with **Load Current**
Power = $V_{DS} * I_{Diode}$ in W
Energy = $\int \text{Power } dt$, = Joule per switch.
Multiply Joule per switch by switch frequency to estimate total Diode Loss
Use V_G and V_{DS} to identify Diode ON period

⊙ = BNC measurement relative shield



⊙ = BNC connection

The Measurement Challenge

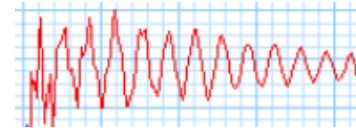
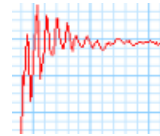
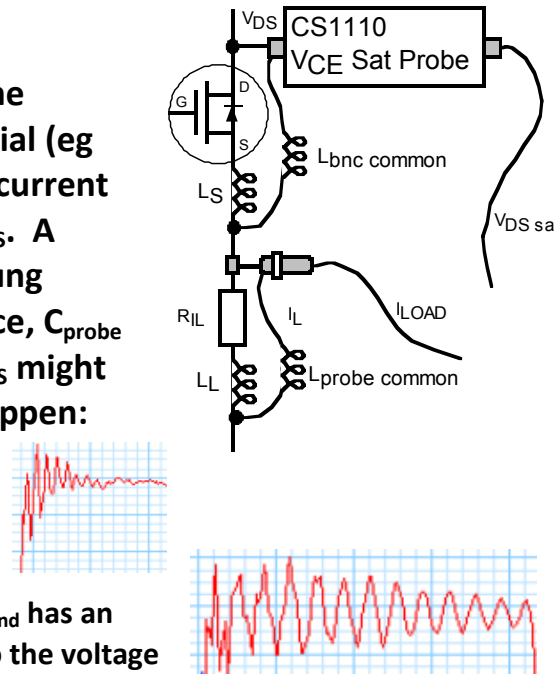
Slew Rate Rejection

The reference common mode voltage (VCM) is attached to and swinging with VOUT. The effective bandwidth of the edge is set by the edge rise or fall time: $BW = \frac{1}{\pi t_r}$
So a SicFet transition of 10ns has an effective bandwidth of $BW = 1/(\pi \cdot 10n) = 32 \text{ MHz}$.

High Inductance connections

The inductance of a piece of wire is about 1nh/mm, so the common connection will have inductance if it is not coaxial (eg $L_{\text{probe common}}$ or L_{PC}). If there is series inductance (eg L_S) in a current path this will add a voltage error to measurements of V_{DS} . A standard ground clip is 140mm long (140nH) while a sprung ground clip is 10mm long (10nH). Probe input capacitance, C_{probe} is 12 pF, and CS448 ground capacitance, C_{ground} is 14pF. L_S might vary 3 - 10nH. Based on these values these bad things happen:

1. The measured signal (eg 15V gate drive) initiates ringing between $L_{\text{probe common}}$ and C_{probe} .
2. The common signal VCM (eg 650V) initiates ringing between L_{PC} and C_{ground} . This combines with (1). L_{PC} has an impedance of about $X = 2 \pi F L = 2 \pi 32M 140n = 28 \text{ Ohm}$. C_{ground} has an impedance of $X = 1/2 \pi F C = 1/(2 \pi 32 \text{ MHz } 14p) = 355 \text{ ohm}$, so the voltage generated across L_{PC} is about $650 \times 28/355 = 51V$, 3x bigger than the 15V being measured! Even with a spring ground clip of 10nH ($X = 2 \text{ ohm}$), you will see $650 \times 2/355 = 3.6V$ added signal.
3. If measuring V_{DS} saturated (say 0.4V) with 10A, $L_S = 10nH$, $V = L di/dt = 10n \cdot 10/10n = 10V$. The inductive voltage totally dominates the V_{DS} .



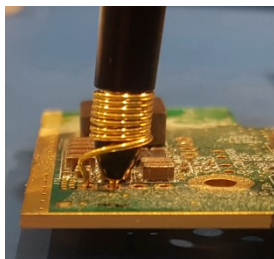
Low inductance four wire connections

This connection method uses four wire connections to avoid measuring L_S , and low inductance connection methods. Coaxial is best. See the next page. Connection methods are:

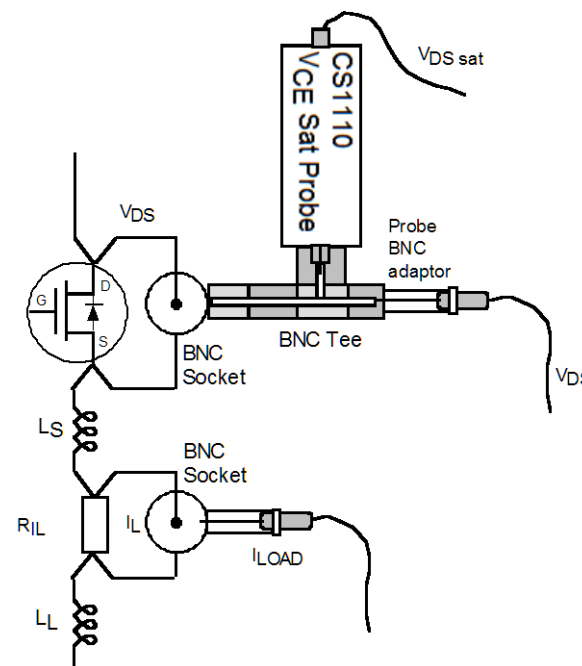
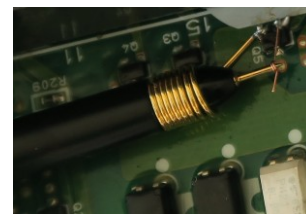
1. Best. Coaxial probe or cable connection. We offer SMA or BNC.



2. OK. Spring ground clip and probe inserted into two vias.



3. Worse. Soldered spring ground clip and soldered wire wrapped around probe tip.



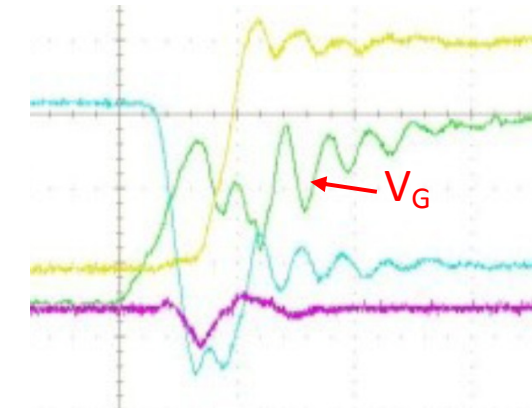
Differential Probe Common Mode Rejection Ratio (CMRR)

You could use a differential probe. The Common Mode Rejection Ratio (CMRR) figure tells you how well the probe rejects common mode slew. Here are CMRR values for PMK Bumble Bee and Tektronix P205A probes, from their data sheets:

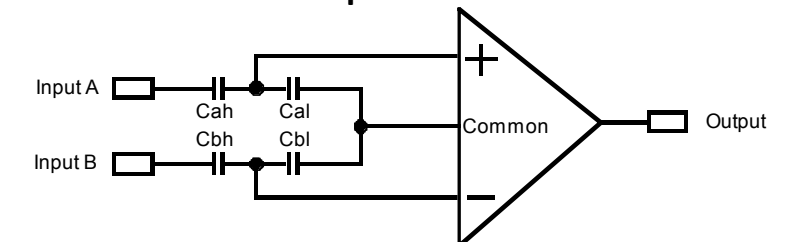
PMK Bumble Bee CMRR	Tektronix P205A CMRR	
DC > 80 dB	DC: >80 dB	
100 kHz > 70 dB	100 kHz: >60 dB	
1 MHz > 62 dB	3.2 MHz: >40 dB	
3.2 MHz > 50 dB	50 MHz: >30 dB	

Let's say 40 dB for 32 MHz, or 1 part in 100. So with the 650V swing, we will see 6.5V of added spurious signal.

It looks something like this:



Why so bad? A Diff Probe uses two capacitive dividers and a differential amplifier:

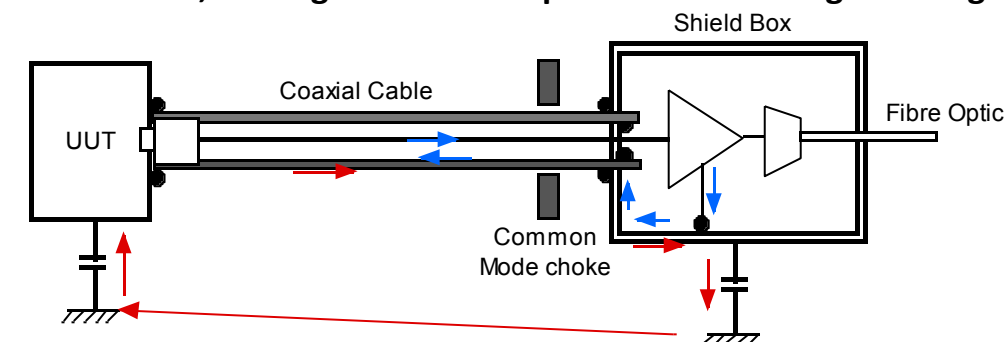


The capacitive divider for each leg is $C_{ah}-C_{al}$ and $C_{bh}-C_{bl}$. The normal input capacitance for each leg is 4 pF. For a 50:1 divider (a common value), C_{ah} will be about 4pF and C_{al} about 200 pF. The common mode rejection is set by the equality of the two dividers. If

the difference between the two dividers is 1 part in 100, then the CMRR will be 40 db, or $20 \log(100)$. For 100dB the match would have to be better than 1 part in 100,000. With $C_{ah} = 4pF$, a match of better than $4/100,000 \text{ pF} = 0.04 \text{ fF}$ is needed. **This is not possible.**

CS448 CMRR

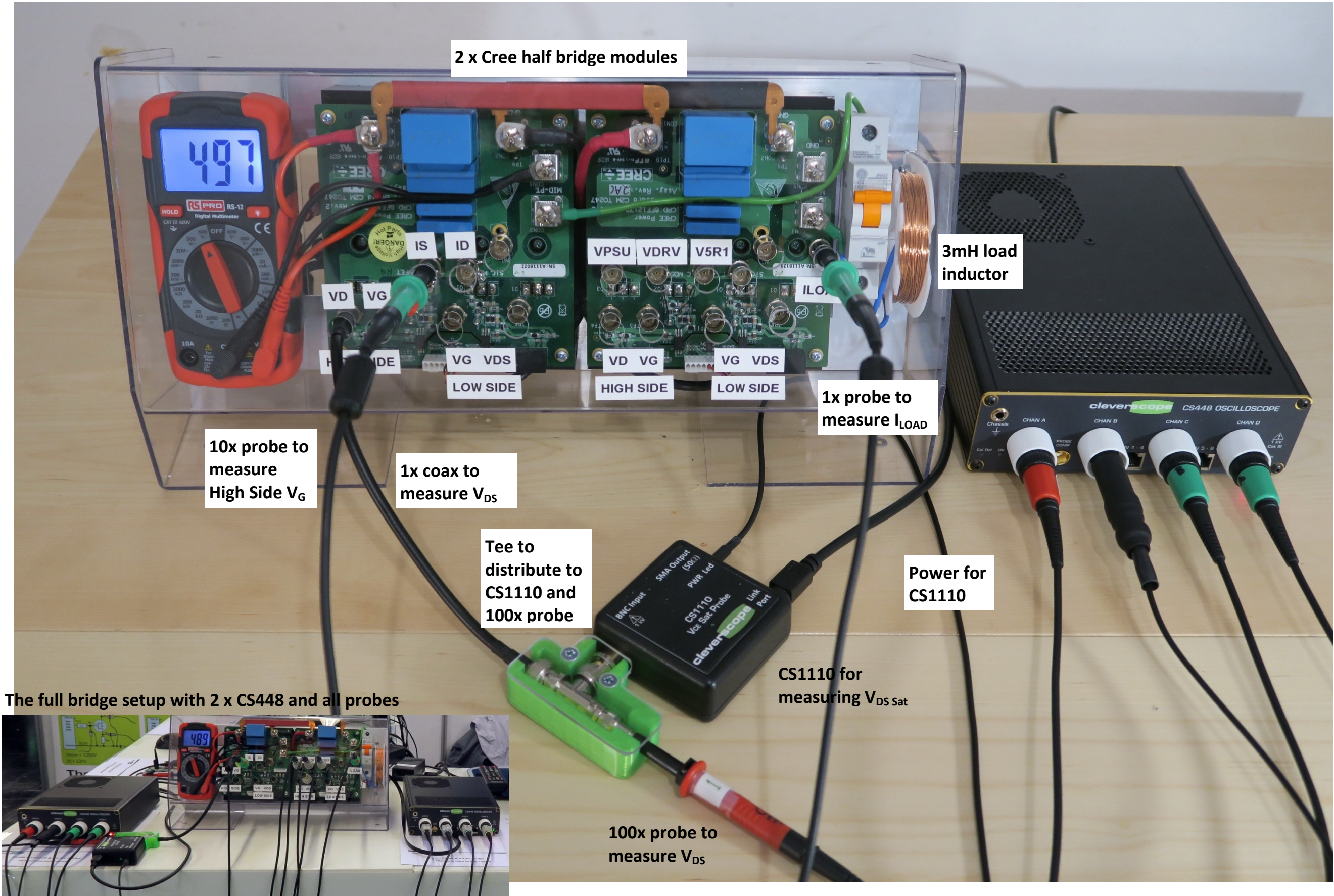
The CS448 has a CMRR of 100 dB at 50 MHz (100,000:1). With the same 650V swing, and a 1x coax connection, you'll see about 7mV of common mode noise. For that, you'll need a coaxial connection, and a good common plane below the signal being measured.



capacitive coupling to the UUT. This large loop has high inductance which can ring with the ground capacitance. Reduce the ring by using a common mode choke to increase the impedance to the common mode current, but not the signal current which sums to zero under the choke.

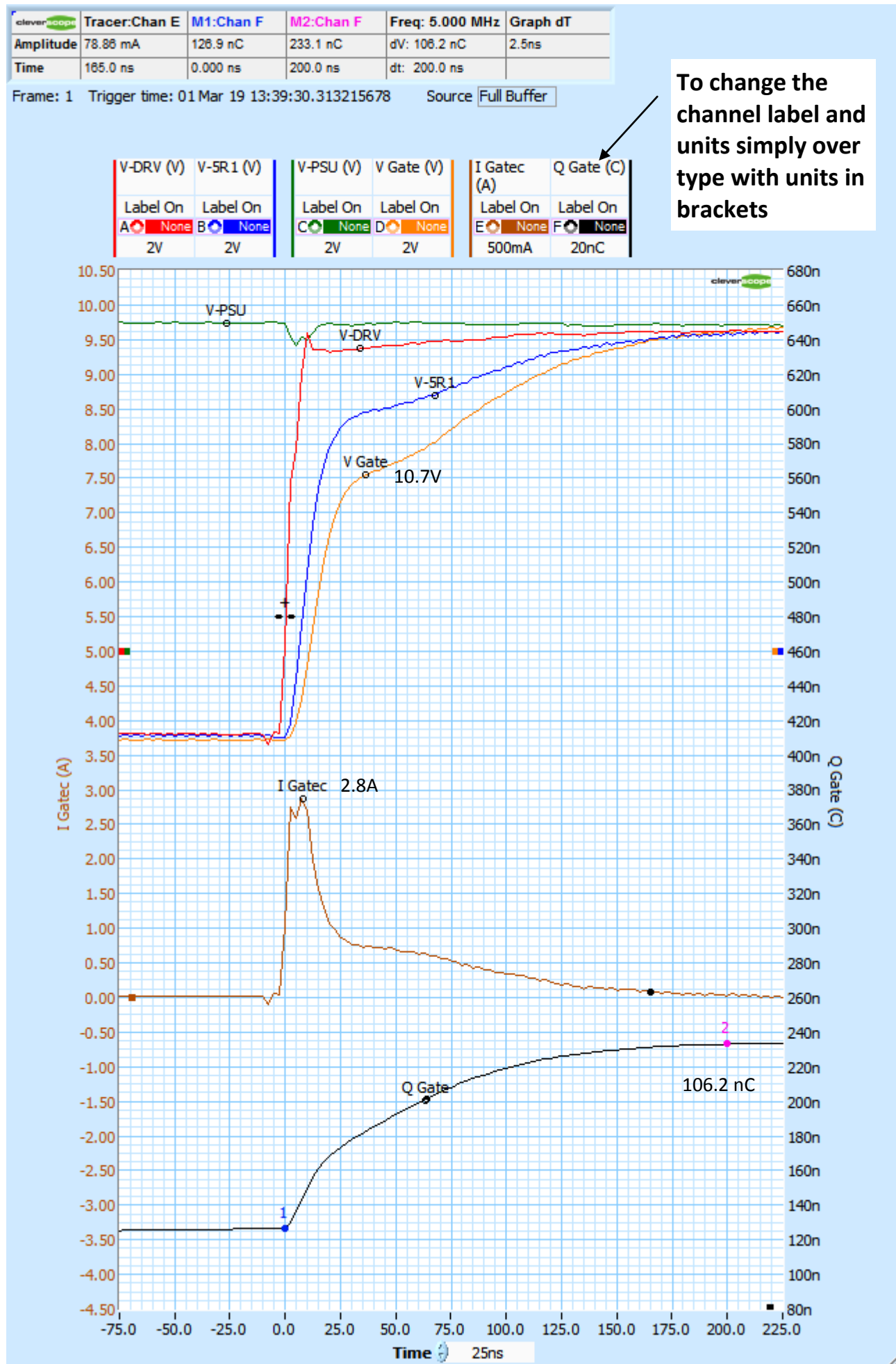
The CS448 is single-ended, and relies on skin effect to separate the signal current (blue) from the common mode current (red). The signal current returns on the inside of the coax shield, while the common mode current flows on the outside of the shield, and returns via

Measurement Setup

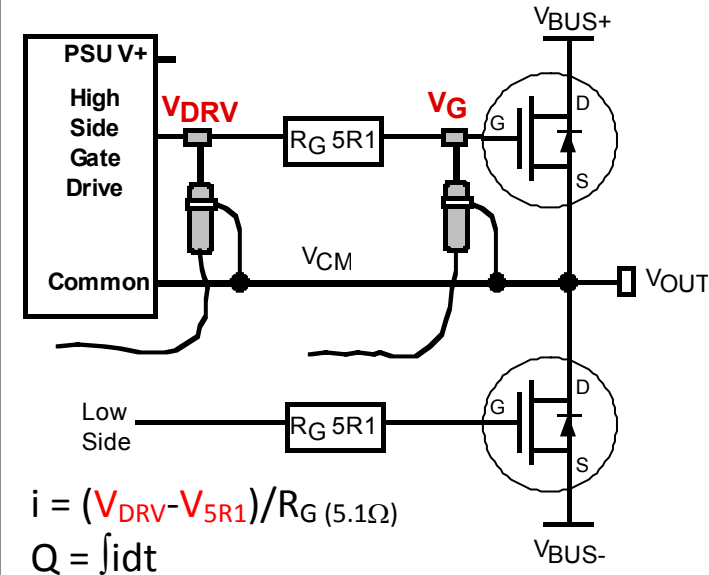


All connections
are BNC coaxial

All measurements
are on the high side.



High Side Gate Voltage and Gate Charge



Measurements

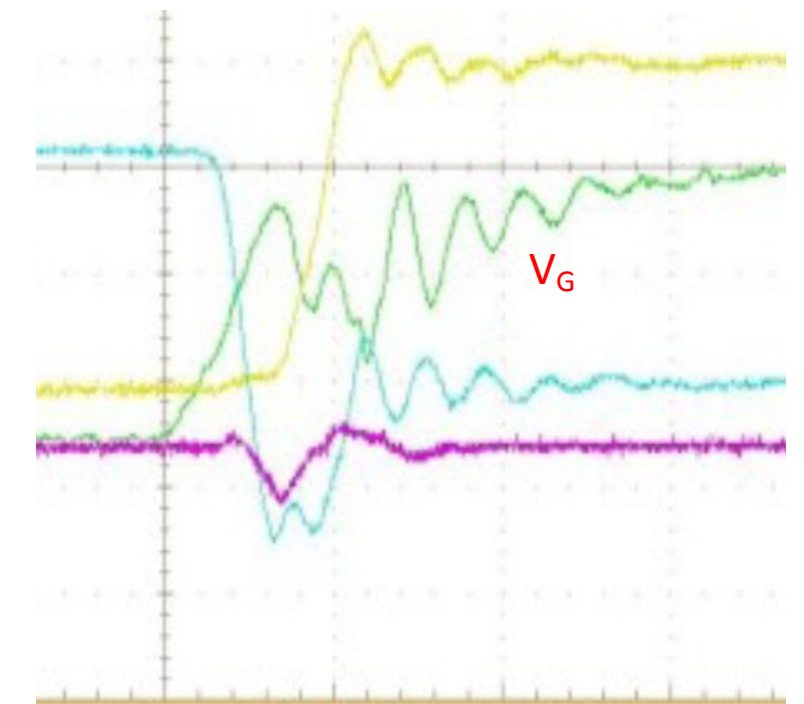
1. V_G has a Miller plateau of about 10.7V. The device is fully on at completion of the Miller plateau.
2. Gate current peaks at 2.8A
3. Total Gate charge is 106.2nC
4. Rise time to the Miller plateau is about 45ns, even though the Drive voltage rise time is around 2ns. The gate drive design has significant R and C induced delay.
5. V_{PSU} dips by 1.34V and recovers in 15ns. Current is 2.7A at the dip, so the power supply output impedance is about 0.5 Ohm.

These measurements are used to verify the design.

Important Note:

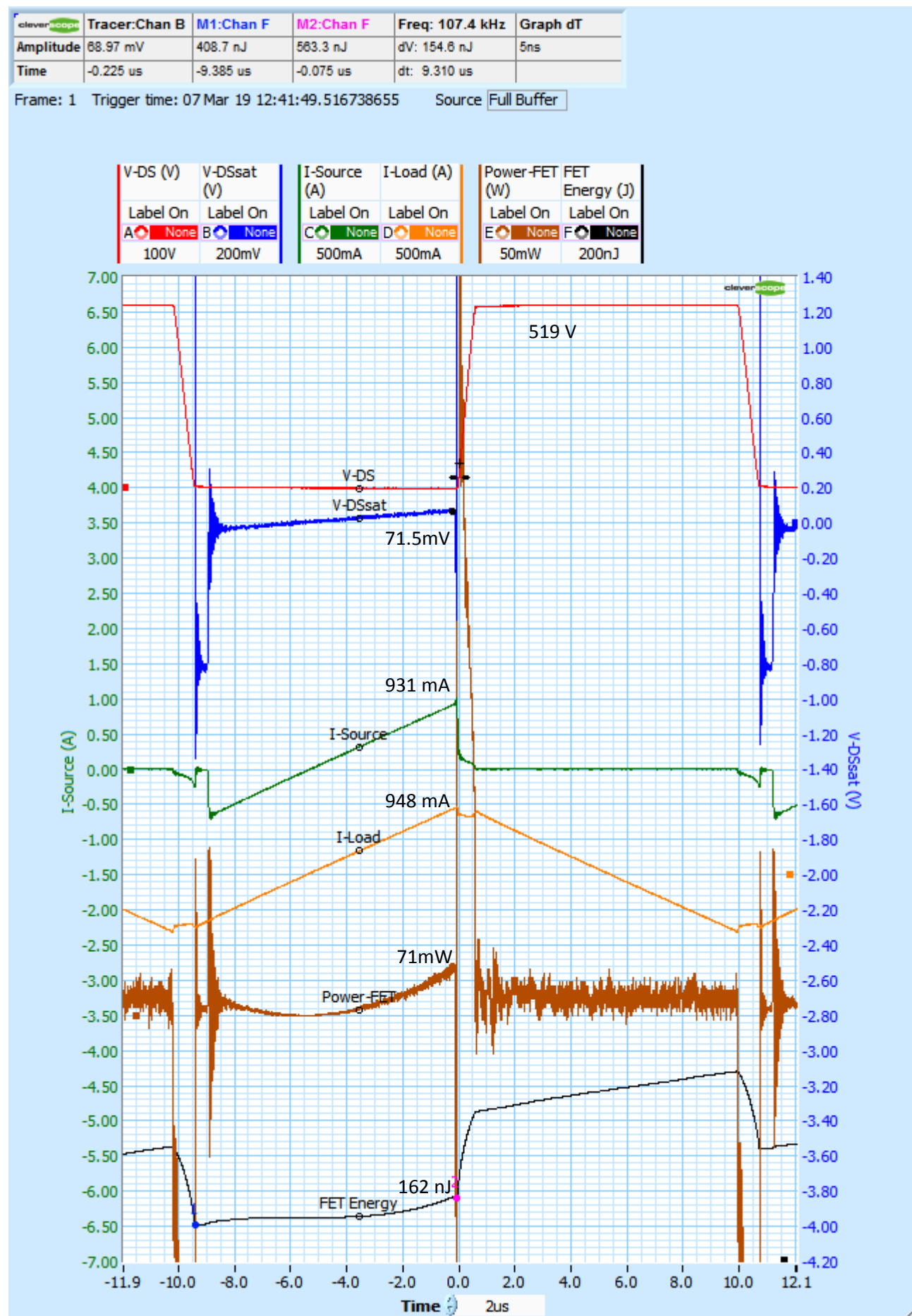
The reference common mode voltage (V_{CM}) is attached to and swinging with V_{OUT} . Very high common mode rejection is needed to make useful measurements of the high side. The CS448 has a CMRR of 100 dB at 50 MHz, meaning it can make clean measurements on a working system.

Comparison with the current state of the art from the Cree SicFET EVM Document: Gate Voltage with a Differential probe (green):

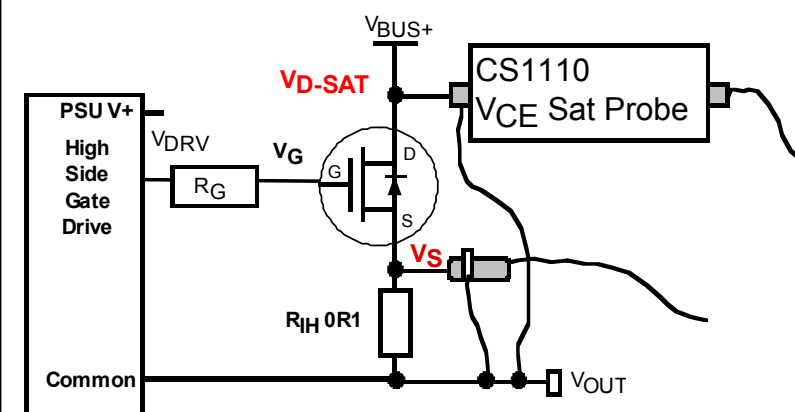


Cree CRD 8FF1217P Eval Board document :

T = 100ns/div, Green = High Side V_{gs} (10V/div), Yellow = Low side V_{ds} (200V/div), Blue = Low side I_d (7.6A/div), Pink = Low side V_{gs} (10V/div)



Conduction Power Loss



$$i = V_S / R_{IH} (0.1\Omega) \quad V_{DS} = V_{D-SAT} - V_S$$

$$P = V_{DS} i$$

$$E = \int P dt$$

Measurements

The Maths display shows the derived values. Values are derived during the high side FET ON period.

- $V_{DS} \text{ (saturated)}$ is derived from $V_{D-SAT} - V_S$ which uses the saturation probe to accurately measure the saturation voltage, which is increasing as the current increases. At the completion of the ON time, V_D is 71.5mV.
- $V_{DS} \text{ (Full)}$ is the voltage the FET is switching. It is 519V. Notice the VCE-SAT Probe is measuring the high side saturation voltage with a common measuring terminal that is swinging between 0 and 519V.
- The FET Current, i , is derived from $V_S / R_{IH} (0.1\Omega)$ and peaks at 931mA with the load being used. It starts off negative, and change to positive as the current in the inductive load reverses. The FET series resistance is $R = V/I = 0.0715/0.931 = 76.8 \text{ m}\Omega$.
- The FET instantaneous power is $P = V_{DS} i$. It peaks at 71mW.
- The FET energy per switch cycle $E = \int P dt$. We measure 162nJ. Assuming 50kHz operation, total energy loss is $50k \times 162nJ = 18mJ/s$. Average power is therefore 8.1 mW.

These measurements are used to verify the design.

Scaling and Maths

Chan C has been scaled so 0.1V measured equals 1A (using the 0R1 current sense resistor), and named I-Source with units of A.

Channel C			
Actual Volts	Desired	Units	Channel C
0.00	0.00	A	I-Source
100.00m	1.00		

The Maths equation Builder equations:

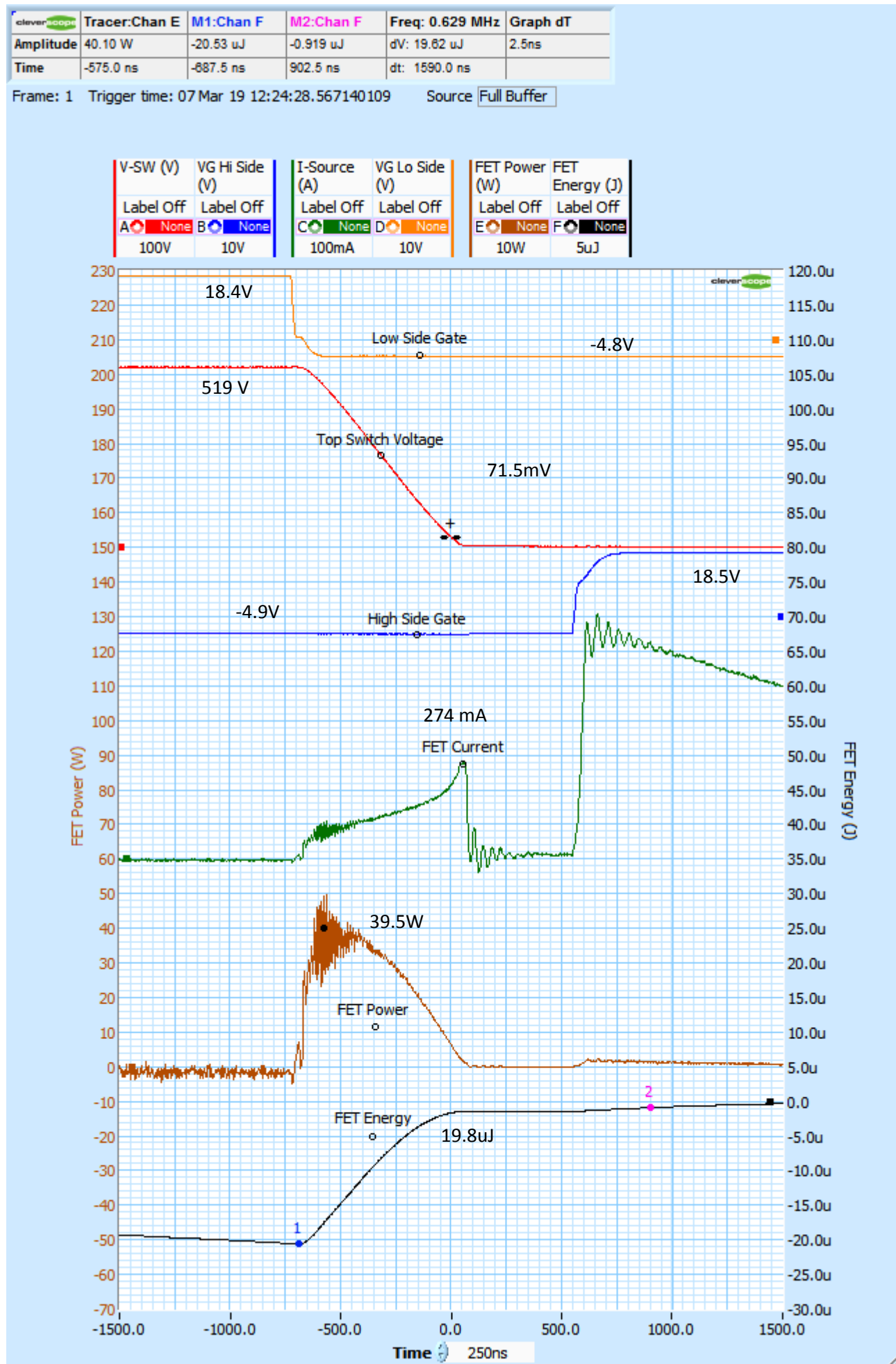
Check Equations	Used?	Process	Destination
$b-(c/10)$	☒	----->	b
$b*c$	☒	----->	e
e	☒	Integral	f

Chan B is V_{D-SAT} . Chan C is I_S and divided by 10 to get it back to volts. The first line calculates V_{DS} . The second line calculates $P = V_{DS} i$. The third line calculates $E = \int P dt$.

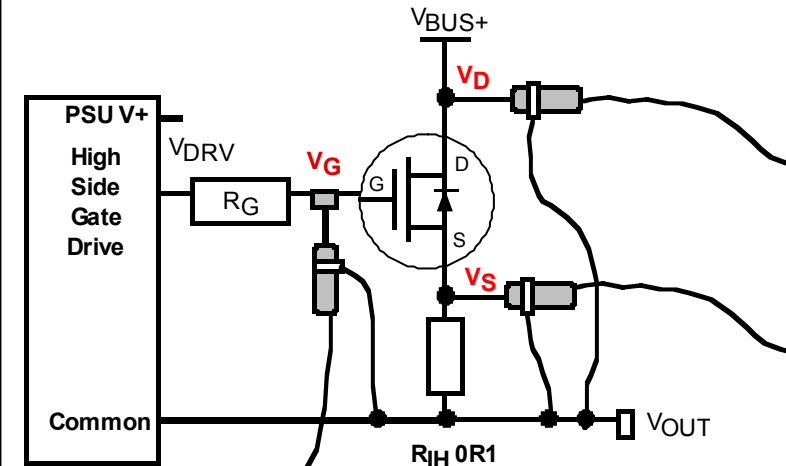
V_{CE-SAT} Probe

The VCE-SAT Probe is used to accurately measure low level voltages (-7 to + 5V) while ignoring input voltages above +5V (1000V maximum). It is isolated from the Link Port (used for power and control), and can be used to measure High Side V_{DS} during saturation.





Switching Power Loss



$$i = V_S / R_{LH} (0.1\Omega) \quad V_{DS} = V_D - V_S$$

$$P = V_{DS} i$$

$$E = \int P dt$$

Measurements

The Maths display shows the derived values. Values are derived during the high side FET switch transition.

1. V_{DS} is derived from $V_D - V_S$ which uses a 100x probe to accurately measure the voltage V_{DS} across the FET, which is decreasing as the device turns on and the current increases. V_{DS} changes from 519V (FET off) to 71.5mV (FET on).
2. The FET/body diode current, i , is derived from $V_S / R_{LH} (0.1\Omega)$ and peaks at 274mA with the load being used.
3. The FET instantaneous power is $P = V_{DS} i$. It peaks at 39.5W.
4. The FET energy per switch cycle $E = \int P dt$. We measure 19.8uJ. Assuming 50kHz operation, total energy loss is $50k \times 19.8uJ = 0.99J/s$. Average power is therefore 1W. The switch losses are much higher than the conduction losses.

These measurements are used to verify the design.

Scaling and Maths

Chan C has been scaled so 0.1V measured equals 1A (using the 0R1 current sense resistor), and named I-Source with units of A.

Channel C		Units	Channel C
Actual Volts	Desired	Name	A
0.00	0.00	I-Source	
100.00m	1.00		
		Units	A
		Name	I-Source

The Maths equation Builder equations:

Check	Equations	Used?	Process	Destination
☒	$a - (c/10)$	☒	----->	a
☒	$c * a$	☒	----->	e
☒	e	☒	Integral	f

Chan A is V_D . Chan C is I_S and divided by 10 to get it back to volts. The first line calculates V_{DS} . The second line calculates $P = V_{DS} i$. The third line calculates $E = \int P dt$.

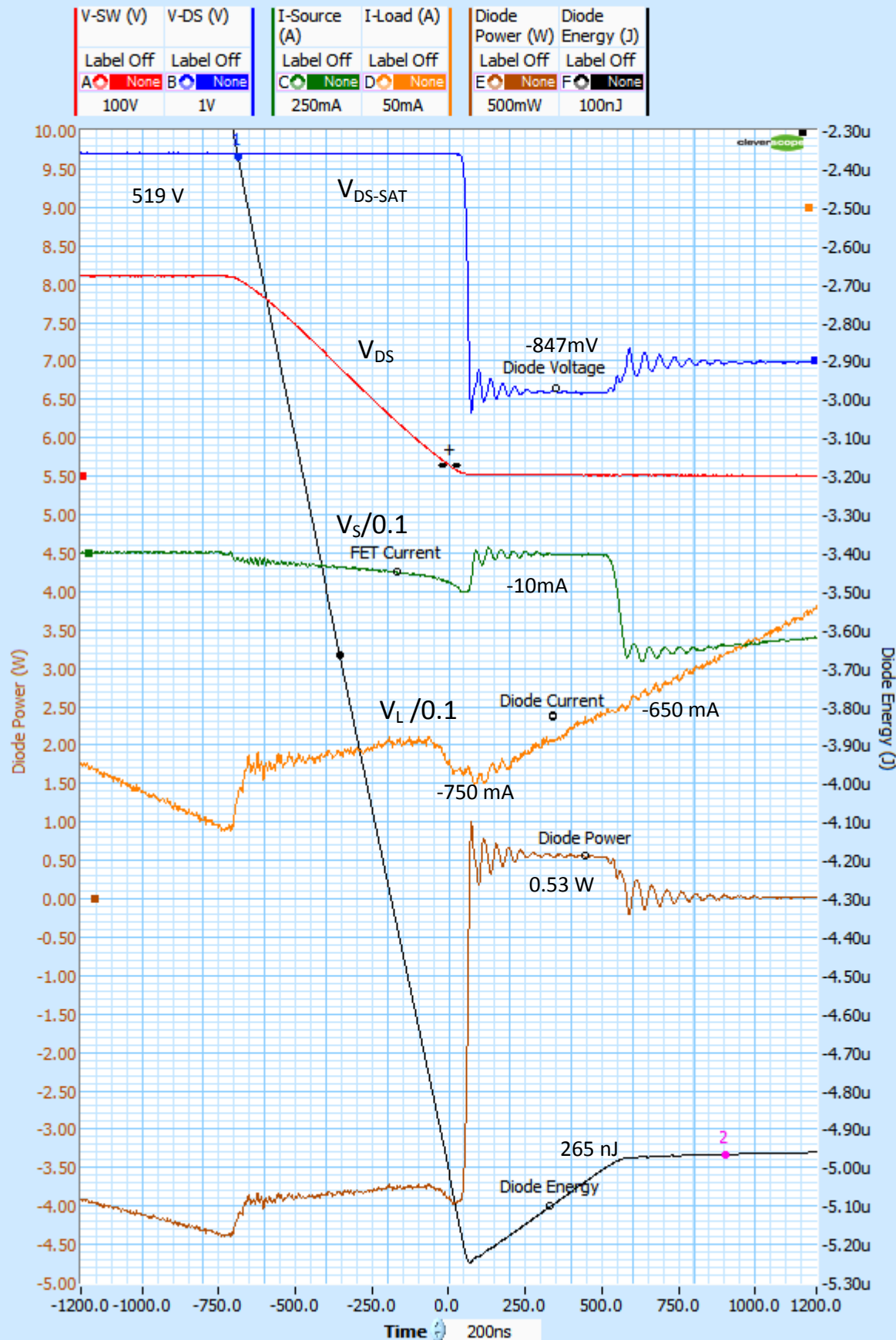
Observations

The Switch loss is high because the inductor current flows to charge the parasitic capacitance in the reverse direction until FET V_{DS} reduces to 0. This current is flowing through the body diode (the FET is off), and starts reducing when the low side FET turns off, and during the dead time. We notice these things:

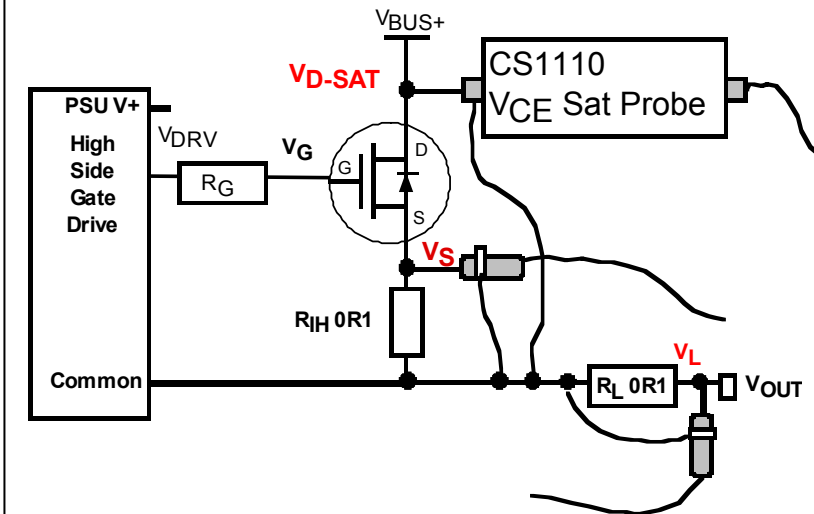
- The high side FET voltage takes 700 ns to reduce to 0.
- The total dead time from low side gate turn off to high side gate turn on is 1277 ns.
- There is initial ringing of 24.9MHz. Further investigation could be used to find the circuit parasitics.

Tracer:Chan F	M1:Chan F	M2:Chan F	Freq: 0.629 MHz	Graph dT
Amplitude	-3884.1 nJ	-2387.1 nJ	-4985.5 nJ	dV: -2598.4 nJ
Time	-355.0 ns	-887.5 ns	902.5 ns	dt: 1590.0 ns

Frame: 1 Trigger time: 07 Mar 19 12:30:30.005417259 Source: Full Buffer



Diode Power Loss



$$V_{DS} = V_{D-SAT} - V_S \quad i_{FET} = V_{D-SAT} / R_{IH} (0.1\Omega)$$

$$I_{Diode} = V_L / R_L (0.1\Omega)$$

$$P_{Diode} = V_{DS} i_{Diode}$$

$$E = \int P dt$$

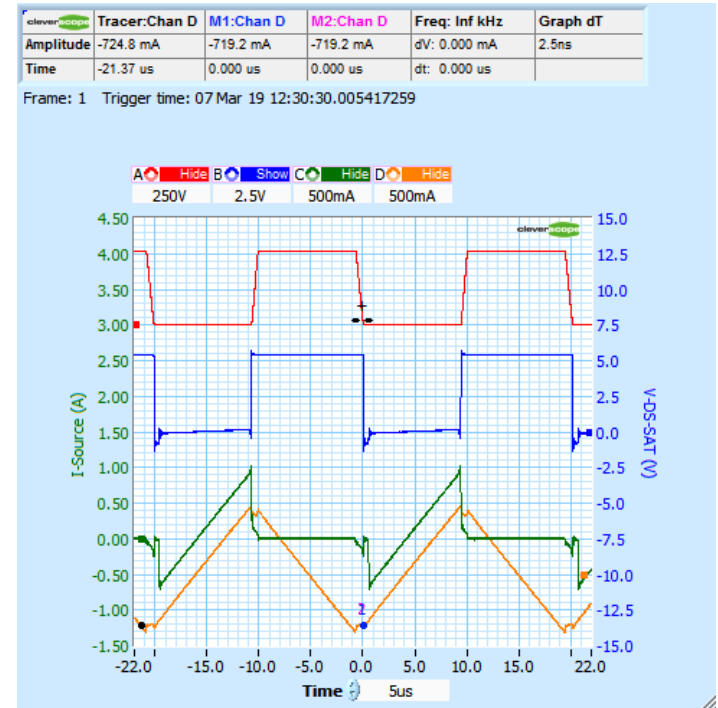
Measurements

The Maths display shows the derived values. Values are derived during the Diode forward on-time which occurs after the low side transistor has turned off and we are in the dead time after the Hi Side V_{DS} has fallen to zero.

- V_{DS} (saturated) is derived from $V_{D-SAT} - V_S$ which uses the saturation probe to accurately measure the diode forward voltage while free-wheeling current flows through it in the dead time. The Diode forward voltage is about 847mV.
- The free-wheeling current, $I_{Diode} = V_L / R_L (0.1\Omega)$, is measured by R_L as it goes to the load.
- The Power dissipated in the diode is therefore $P_{Diode} = V_{DS} i_{Diode}$. Instantaneous power is about 0.53W.
- The Energy dissipated in the diode per switch cycle is the integral of the instantaneous power, $E = \int P dt$. We measure 265nJ. With 50k transitions per second, this is 13mJ/s, or 13 mW.

These measurements are used to verify the design.

The point at which the Diode is on needs to be identified. This is the point at which the diode is forward biased at the end of the dead time.



Scaling and Maths

The channels measuring V_S and V_L have been scaled so 0.1V measured equals 1A.

The Maths equation Builder equations:

Check	Equations	Used?	Process	Destination
☒	$b-(c/10)$	☒	----->	b
☒	$b*d$	☒	----->	e
☒	e	☒	Integral	f

Chan B is V_{D-SAT} . Chan C is I_S and divided by 10 to get it back to volts. The first line calculates V_{DS} .

Chan D is I_{Diode} , and the second line calculates $P = V_{DS} i_{Diode}$.

The third line calculates $E = \int P dt$

CS448/CS328A Additional Capabilities

Frequency Response Analysis

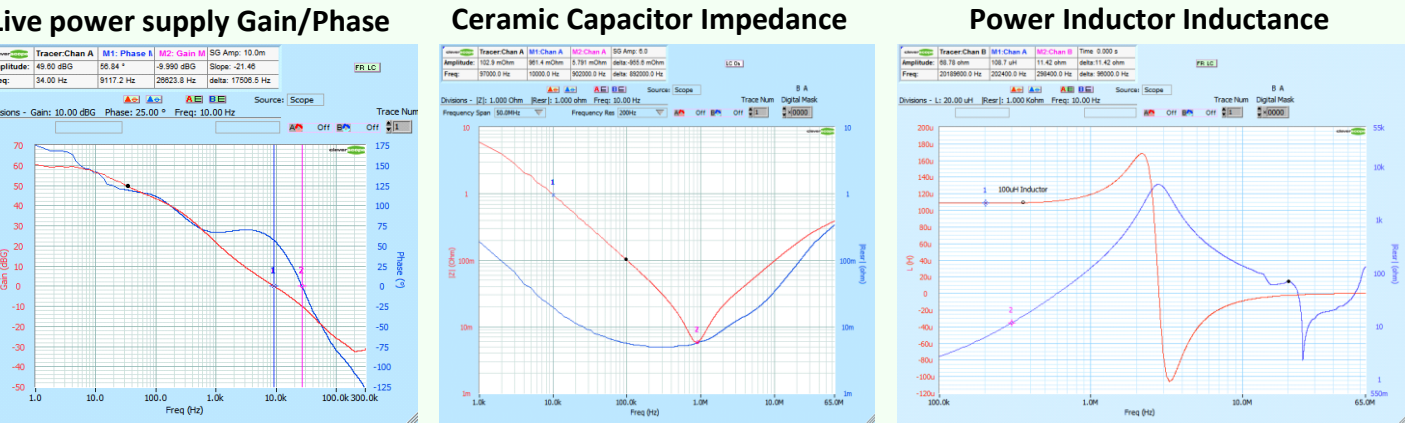
Passive component

- RMS or Power response
- Gain/Phase
- Impedance / Phase, Q, Resr
- Inductance / Phase, Q, Resr
- Capacitance / Phase, DF, Resr

Easily analyse powered active or passive networks to meet design intent.

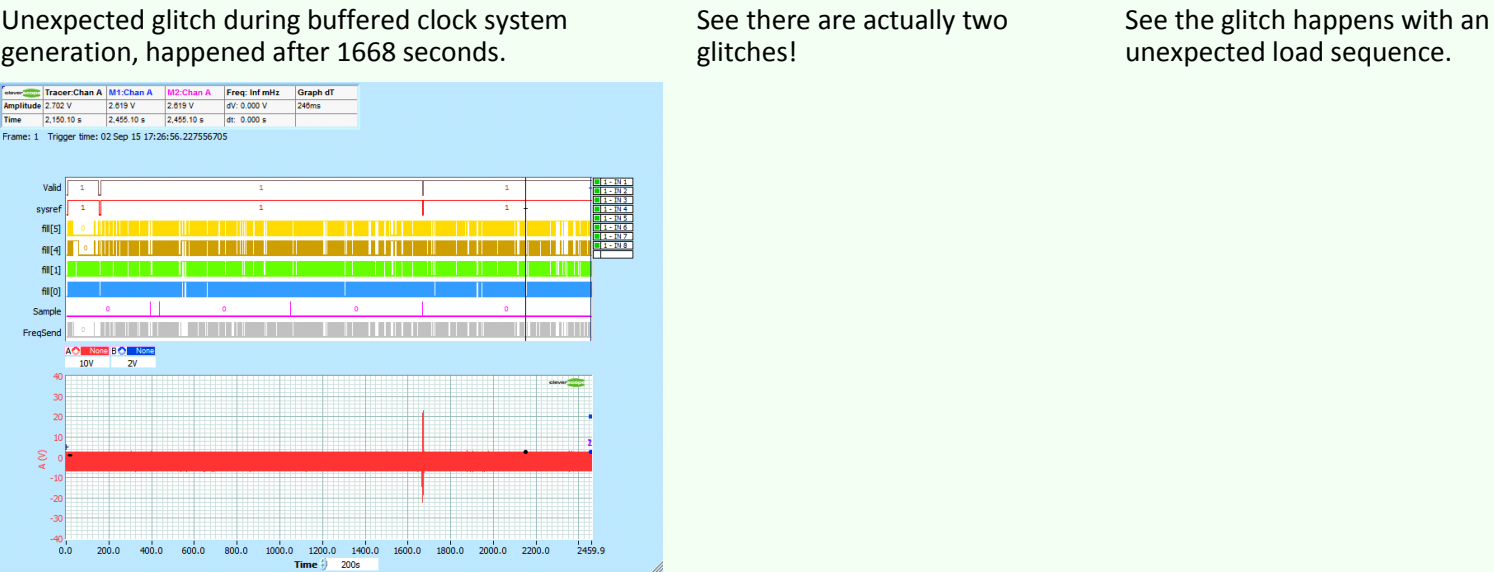
Powered Op Amp or Power Supply

- Gain/Phase
- Input Impedance / Phase
- Output Impedance / Phase
- Power Supply Rejection Ratio



Streaming

Easily save very long captures (secs to months) to disk with as fine as 200ns resolution, including 2ns peak capture. Find unexpected features, zoom in quickly to investigate.



Standard Capabilities

- Symbolic Maths with live Matlab link (with 10 lines of equations).
- Spectrum Analyser with settable Bandwidth and Resolution
- Protocol Decoding
- Signal Information Display with 47 functions and logging to Excel
- Tracking display for very fast examination of fine detail
- Mixed Signal triggering with dual triggers including counting and period

Bart Schroder
Ken Henderson
www.cleverscope.com

Maths Equation Builder Equations

The CS448 application uses the Maths Equation Builder to derive plots for Power, Energy, Current etc. You can have up to 10 maths equations, including symbolic, conditional and function components.

1. High Side Gate Current and Charge

Maths Equation Builder			
Check	Equations	Used?	Process Destination
☒	(a-b)/5.1	☒	-----> e
☒	e	☒	Integral f
☒		☒	-----> c
☒		☒	-----> d
☒		☒	-----> d
☒		☒	-----> a
☒		☒	-----> a
☒		☒	-----> a
☒		☒	-----> a

2. Conduction Power Loss

Check	Equations	Used?	Process Destination
☒	b-(c/10)	☒	-----> b
☒	b*c	☒	-----> e
☒	e	☒	Integral f

3. Switching Loss

Maths Equation Builder			
Check	Equations	Used?	Process Destination
☒	a-(c/10)	☒	-----> a
☒	c*a	☒	-----> e
☒	e	☒	Integral f

4. Diode Loss

Maths Equation Builder			
Check	Equations	Used?	Process Destination
☒	b-(c/10)	☒	-----> b
☒	b*d	☒	-----> e
☒	e	☒	Integral f

Unit names and scaling

Scale input signals simply by specifying the input value, and the desired output value. Set the names of Units, and the Unit by typing them in. They will be used everywhere.

Analog Names and Units

Convert Volts to custom units of your choice

		Channel A	Channel B	Channel C	Channel D
		Actual Volts	Desired	Actual Volts	Desired
Acquired Samples	Pt. 1	0.00	0.00	0.00	0.00
	Pt. 2	1.00	1.00	100.00m	1.00
Frequency Spectrum	Pt. 1	0.00	0.00	0.00	0.00
	Pt. 2	1.00	1.00	1.00	1.00
	Reference voltage for 0dB (log) or for 1W	1.00	1.00	1.00	1.00

Custom Names and Units

	X Axis	Channel A	Channel B	Channel C	Channel D
Time and Tracking Graph	Units Name	s Time	V V-SW	V V-DS	A I-Source
Maths Graph	Units Name	s Time	V V-SW	V V-DS	A I-Source

CS448 Application

Notice these features:

1. Axes are labeled with the signal name, and are in useful units, and include grid lines with numeric values.
2. Derived values such as current and charge are easily generated with the maths equation builder.
3. The graph can be annotated for ease of understanding.
4. Markers are used for measurements.
5. Inserting the graph into a document is simply Ctrl C + Ctrl V.